HBV, HCV co-infection among blood donors in Nigeria

Hepatitis B Virus (HBV) and *Hepatitis C Virus* (HCV) are major pathogens responsible for viral hepatitis in the tropics, especially in Nigeria, causing both acute and chronic liver diseases such as hepatocellular carcinoma (HCC) and liver cirrhosis.[1,2] Transmission of HCV predominantly takes place as a result of blood transfusion and exposure to blood derivatives.[3]

To evaluate the risk of post transfusion-associated viral hepatitis infection among Nigerians, the HBsAg and anti-HCV antibody in the plasma samples from blood donors was assayed.

The study surveyed 624 blood donors froma teaching hospital in the south western part of Nigeria and some privately owned blood banks. The samples were tested for the presence of HBsAg and anti-HCV using a qualitative technique based on the principle of antigen-antibody reaction (Clinotech diagnostics and pharmaceuticals, Inc. 11-12240 horse shoe way Richmond, Canada.) Results show that seroprevalence of hepatitis B surface antigen, antibodies to hepatitis C virus and co-infection of HBV/HCV in the tested subjects were 13.5, 3.0 and 1.1% respectively.

It was also shown that HBsAg and anti-HCV rates were higher among males than their female counterparts while co-infection among the two sexes was not different. Among the age groups, subjects aged 58 or more had the highest seroprevalence of 25.0% for HBsAg and anti-HCV while the difference between HBsAg and anti-HCV in male and female was not statistically significant.

In 2002, the World Health Organization (WHO) estimated that most African countries had no national blood policy, and some countries with policies encouraged volunteer donor recruitment.[4] This review also estimated that 25% of the blood was untested for anti-HIV, while, HBsAg was screened in 50% of donors or donations and only 19% were screened for anti-HCV. In Nigeria, from 2005, the national blood policy directed that all donated blood should be screened for HBV, HIV, HCV and syphilis. However, this (especially screening for HCV) has not been fully implemented in all private blood banks.

This study has shown that both HBV and HCV infection are endemic in Nigeria while HBV infections seem to be more endemic, Our findings in similarity with those of Allain *et al*.[5] whichidentified that anti-HCV is not been screened for in most blood banks in African countries. We therefore recommend the need for regular screening of blood not only for HBsAg, as in most blood banks, but also for anti-HCV the cost of the screening test kit not withstanding; whether in the government or private owned blood banks, to minimize risk of hepatitis B and C virus

transmission. The government may also see to subsidize screening test kits to make it affordable.

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